Concept Project Description XR (10EC)

Project Guide: Immersive Intelligent Design

Course: XR (10 EC)

Duration: 10 weeks (Delivery & Demo in week 10)

Theme: Designing XR-Learning Experiences with Integrated AI

Technical project results are rarely achieved in isolation, they are developed through teamwork, communication, planning, and continuous improvement. The project is therefore designed as one integrated learning environment to develop and assess two performance domains: technical competence and professional attitude. The two dimensions are assessed using different rubrics to evaluate both what you can do technically (PI6) and how you work professionally (PI9).

1. The Project

In this block, you will act as a specialized Ed-Tech Agency (Educational Technology). Your team is an expert in combining Spatial Computing (XR) and Artificial Intelligence.

You are hired by a domain expert (your client). This expert has a lot of knowledge but is looking for an effective way to teach this knowledge to others. The question is not simply "build an app," but "design a learning experience."

The goal:

1. You will develop an XR Training in Unity.

   AI must be integrated into this experience. The AI is not just a gimmick; it must be an essential part of the app (for example, a smart tutor, a dynamic opponent, or a voice-controlled guide).

   Your team is responsible for the content and the teaching method. You must formulate research questions. By using desk research (about learning in XR) and user research (about the target group), you will determine the story and structure of the training.

2. You will demonstrate professional attitude

Your professional attitude is assessed throughout the project. You are expected to participate actively in project and lab activities and to demonstrate the professional behaviours described in the PI9 rubric. Your tutors observe your professional behaviour and contribution, while your **portfolio** provides evidence of your preparation, work, feedback, reflections and development.

There is no formal attendance requirement; however, you need to participate in sufficient lab sessions for your professional attitude to be observed and assessed validly. If there is insufficient evidence of your professional behaviour and development, PI9 cannot be assessed as sufficient.

2. The Three Research Tracks

You will work on one of the topics below. The experts are your starting point to understand the problem and the context.

Track A (Class A): Digital Twins & Big Science (Einstein Telescope)

Context: The Einstein Telescope is a massive underground measurement instrument. It is designed to observe the universe by detecting gravitational waves. The underlying physics and geophysical concepts are highly complex. Digital Twinning is used to make invisible data visible. The goal is sense-making of these concepts and information

Client: Josi Menezes da Silva (Senior Researcher Knowledge Centre Data Intelligence).

Track B (Class B): Applied Engineering for the Einstein Telescope

Context: The Einstein Telescope is not just a scientific instrument; it is an enormous engineering challenge. The engineering students of today are the future professionals who will build, operate, and maintain this massive underground facility. Working in this environment requires mastering complex procedures and interacting with highly sensitive equipment. The research goal is to design an experiential XR simulation where these future engineers can safely practice working inside the telescope.

Client: Agnes Berendsen & Jesse van Dongen (Project lead Einstein & Lecturer / Researcher Engineering).

Track C (Class C): Healthcare & Wellbeing (Project CARAI)

Context: The CARAI project wants to make healthcare professionals "AI-ready". A key objective here is to significantly improve the AI literacy of healthcare professionals. There is a need for training that bridges the gap between daily care work and abstract computer algorithms, ensuring that healthcare workers understand AI sufficiently to trust and use it effectively in their daily routines.

Client: Marcel Schmitz (Senior Researcher Knowledge Centre Data Intelligence).

Core Design Principles

As a professional Ed-Tech Agency, your team does not just build applications randomly; you design with purpose. To ensure that your XR training is effective, accessible, and ethically sound, you must adhere to a set of mandatory design principles.

We have established a baseline framework consisting of **General Principles** (applicable to all projects) and **Track-Specific Principles** (tailored to the unique context of your client).

1. General Principles

All teams must strictly follow these overarching guidelines when designing their learning experience:

- Research and select one or multiple learning theories suited to your user and goals. Extract their specific instructional principles and apply them directly to your overall design and mechanics of your XR application
- Clearly communicate the learning goals to the user upfront and embed formative assessments directly within the XR application to guide and evaluate their progress.
- Your XR application must mitigate adverse health effects
- Integrate AI as a functional component of the XR experience.
- Design the XR application to run effectively either fully standalone on the headset or hard-tethered to a PC.

2. Track-Specific Design Principles

Depending on the track you have chosen, your solution must also integrate the following specific requirements:

Track A: Digital Twins & Big Science (Einstein Telescope - Physics)

- Utilize spatial computing to make invisible data (such as gravitational waves) physically manipulable. The interaction design must allow users to grab, scale, and manipulate abstract data representations, translating complex physics into intuitive, tangible experiences.
- To manage the high cognitive load associated with complex geophysical concepts, users must be able to interactively filter and reveal data layers within the Digital Twin. Interactions should allow smooth transitions from a macro-level overview to micro-level scientific details.
- *The track-specific principles above represent only your baseline. Through your desk research (learning and XR interaction theories) and user research (target interviews), your team must discover, define, and validate at least 2–3*

additional interaction and functional requirements tailored to your specific user needs.

Track B: Applied Engineering (Einstein Telescope - Engineering)

- The affordances of virtual tools and sensitive equipment must closely mimic real-world physical constraints and interactions (e.g., precise turning, calibrating, and handling). This ensures realistic motor-skill development and valid transfer of engineering skills.
- Leverage the XR environment to provide clear, immediate visual and haptic feedback when complex procedures are performed incorrectly. The design must allow future engineers to safely experience the consequences of mishandling equipment without real-world risks.
- *The track-specific principles above represent only your baseline. Through your desk research (learning and XR interaction theories) and user research (target interviews), your team must discover, define, and validate at least 2–3 additional interaction and functional requirements tailored to your specific user needs.*

Track C: Healthcare & Wellbeing (CARAI)

- Progressively introduce abstract AI concepts through intuitive, step-by-step interactive tasks. The XR experience should start with recognizable daily care routines and slowly layer in complexity, preventing cognitive overload.
- Design interactions that visually explain how the AI makes decisions within familiar, everyday healthcare scenarios. The system must provide transparent, on-demand explanations for its output to foster trust and practical understanding among healthcare professionals.
- *The track-specific principles above represent only your baseline. Through your desk research (learning and XR interaction theories) and user research (target interviews), your team must discover, define, and validate at least 2–3 additional interaction and functional requirements tailored to your specific user needs.*

3. Enriching the Framework: Your own research-driven principles

Important: The principles listed above form your *baseline*, not your finish line.

As part of your **Phase 1: Research & Definition**, you are required to enrich this list with your own set of design principles. Based on your Literature Study (Desk Research) and User Research, you must distill additional principles that are specifically tailored to your exact target audience and learning goals.

Instruction: You must explicitly document and justify your self-defined design principles in the **Design & Realization** chapter of your final Research Report, showing clearly how they stem directly from your research findings.

3. The Approach: Research-Based Design

You will follow a cycle where your research guides your design and technical work.

Phase 1: Research & Definition (Week 1-3)

Do not start building yet. First, investigate the problem.

- **Defining the Problem:** After interviewing the expert, you formulate your main research question.
- **Literature Study (Desk Research):** Read the theory. What are proven methods for learning and training in XR? Interaction Design in XR?
- **User Research:** Who is the user? What do they already know? What do they find difficult or scary?

Educational Note: A common mistake is skipping the literature. If you don't know how people learn in VR, you cannot build a good VR training. Use the theory to make smart design choices.

Phase 2: Design & Development (Week 4-8)

Based on your research, you design and build the solution.

- **Narrative Design (The Script):** You write the script of the training. The scenarios and the role of the AI are based on your research.
- **Building in Unity:** You build the prototype in Unity.
- **AI Integration:** You connect an LLM (Language Model) inside the application. The AI makes the app adaptive (e.g., generating unique dialogues or explaining data on request).

Phase 3: Validation (Week 9-10)

- **Testing:** You test the prototype with real users. Does the interaction work? Did they learn what they were supposed to learn?
- **Conclusion:** You answer your research questions in the final report.

4. Deliverables

The project consists of three deadlines, which are due in week 3, 6, and 9. Submission buttons can be found on Moodle. In the PI6 and PI9 rubric, you can see what is assessed

for PI6 (e.g., the report) and what is part of the PI9 assessment (e.g., whether the assignment was turned in on time).

Week 3:

- Report part 1 (Title page, introduction, theoretical framework)
- Collaboration contract
- User research focus group interview recording
- Co-Creation results

Week 6:

- Report part 2 (Part 1 + Methodology)

Week 9:

- Full report
- Product deliverables
- PI9 Portfolio

These final products must be delivered, passing grades are based on the PI6 rubric:

1. **The working Unity Prototype:** Executable app with AI integration that runs on a headset either standalone or via cable.
2. **The Research Report:** (Portfolio).
3. **Video Demo:** A video of your application with a maximum length of 15 minutes.

Quality Standard: Applying Professional Methodologies

Building on your prior courses (Intro to HCDI, Design of Interactive Systems, Data Viz, Development), we expect you to explicitly apply and document the following professional methodologies in your product and portfolio:

- **User Research & Analysis:** You must gather qualitative data using **appropriate methods (e.g. Semi-structured Interviews and Observations).** You represent your analysis using Affinity Diagrams (and similar thematic analysis methods) to create data-driven Personas and User Journey Maps.
- **Design & Prototyping:** Follow an iterative User-Centered Design (UCD) process. We expect to see the evolution of your concept through Sketching, Wireflows, and Lo-Fi/Hi-Fi Prototyping. Justify your interaction design choices using HCI principles (e.g., Gestalt, Affordances).
- **Validation:** Verify your solution using structured Usability Testing (e.g., Think Aloud protocol) and analyze the results to validate your design decisions.
- **Technical Quality:** Your C# scripts must meet professional standards:

- **Clean Code:** Write readable code. Use clear names for variables and functions (e.g., `ToggleNightMode()` instead of `DoStuff()`).
- **Structure:** Keep your project folder neatly organized (folders for Scripts, Materials, Prefabs, Scenes).
- **Error Free:** The application must run without errors or warnings in the Console.

In addition, you are expected to maintain evidence throughout the block of your professional attitude in your portfolio for the assessment of PI9. It is not enough to simply fill out your portfolio at the end; this must be done throughout weeks 1 through 9.

Structure & Contents of the Interim Presentations (Week 6)

Duration: 8 minutes **Audience:** Product Owners, Lecturers, and Peers

- **Introduction:** Briefly introduce your team, the chosen track, and the central problem you are addressing.
- **Research Questions & Hypotheses:** Clearly share the main research questions and hypotheses that are guiding your project.
- **Initial Research Results:** Present the preliminary findings from your literature study and user studies.
- **The Project Concept:** Present the core XR project concept you have developed based on your research.

Structure of the report

The report is not a diary. It is a professional document where you justify your design choices using the standards mentioned above. The report must contain the following chapters:

1. Title Page

- Project title, student names, date, class/group, client name.

2. Management Summary/Abstract

- A short summary of the problem, the solution (your concept), and the main test results.

3. Technology statement (the use of AI/LLM Tools)

You are required to include a brief reflection on your use of AI tools, specifically Large Language Models (LLMs) such as ChatGPT, Gemini, or Copilot.

This reflection must include the following elements:

a. Usage

Indicate whether you used an LLM. If so, describe **how** you used the tool (e.g., for idea generation, rephrasing text, producing examples, drafting interview questions, structuring sections, etc.).

b. Degree of reliance

Explain to what extent the LLM's output was directly used, adapted, or rewritten. Specify which parts of the work were formulated or significantly modified by you.

c. Critical reflection

Briefly reflect on how you assessed the quality, accuracy, and relevance of the AI-generated content. Mention any limitations or risks you encountered when using the tool.

This reflection is a mandatory component of your submission and will be considered in the evaluation of your process awareness, academic integrity, and transparency.

Important: The use of LLMs is permitted also for coding, provided it is applied responsibly and transparently.

4. Introduction

- Description of the context (CARAI, ET, or Society).
- Why this project is needed supported by societal relevance.
- The central problem.
- The Research Questions (Main question and sub-questions).

5. Theoretical Framework / Literature Study

Show that your design is based on existing knowledge, using existing sources.

- Research on Learning in XR: Which theories about learning and interaction (e.g., cognitive load, gamification) did you use?
- Research on Human-AI Interaction: How do you design a natural conversation with an AI?
- Citing sources is important and mandatory here.

6. Methodology

- How did you work? (e.g., Design Thinking).
- How did you set up your research?
- How did you do User Research? (Interviews, observations).
- How did you do the Validation (Usability Test)? (The plan, the participants, what you measured).
- Describe the target group & participants.

7. Results

It has three parts:

- **6.1 User Research Results:** Analysis of the interviews. Describe the insights (Needs & Pains) and show the Personas and User Journeys. This is the foundation of your design.
- **6.2 Design & Realization:** Translation from research to product according to General, Track-specific and Self-defined design principles. Describe the Narrative (the script), show Storyboards/Wireflows, and explain the technical setup (Unity & AI code). Explicitly mention applied HCI/Design principles.
- **6.3 Validation Results:** The outcomes of the user tests. What went well? Where did users get stuck? Were the learning goals achieved?

8. Discussion and Conclusion

- Answer the research questions based on the results.
- Reflect on the limitations of your prototype and give suggestions for future work.
- Reflect on the process and teamwork.

9. Recommendations

- Advice for the client on next steps.

10. References

- List of all sources used (APA style) - examples.
- For every source describe the following information:

How and where did you find it?	Specify: the used keywords and database you searched, or through which refence you got to this source.
Design & participants	Specify the scientific methods that this study used, as well as a description of the participant group.
Relevance in max. 2 sentences.	Describe why this reference was important for your project.

11. Appendices

- Interview notes.
- Test plans and raw anonymized data.
- Technical documentation (code snippets, prompt engineering logs).